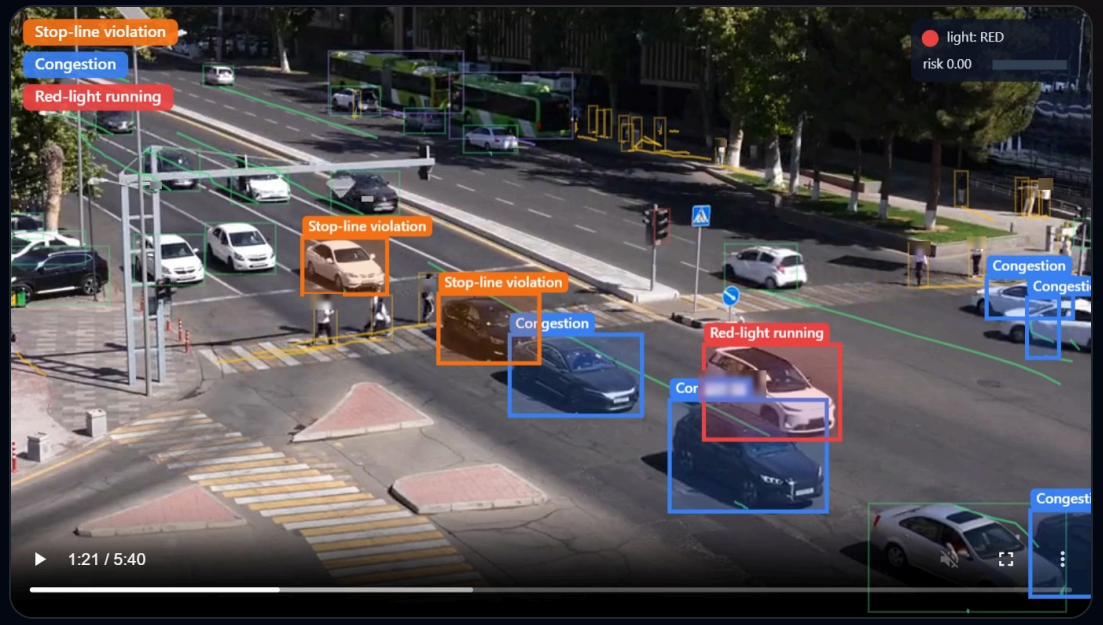
Mode: Startup

Do’ppiX SafeJunction

Turn any traffic camera into a road-safety auditor

It finds red-light running, drivers not yielding to pedestrians, wrong-way driving and near misses on the CCTV a city already has, and tells the traffic engineer what to change, before the crash.



Eldor Musayev · Shohruh Gulmirodov · Ibodulla Obidjonov | WIUT Hackathon 2026 Final · Computer Vision

PROBLEM AND USER

Cities learn which junction is dangerous from crash reports. That is too late

Evidence: one Tashkent intersection, 18.4 min of video, labeled by us

What we saw	Count	Per hour
Pedestrians on the roadway, off the crossing	16	52 / h
Vehicles not yielding to pedestrians on the crossing	9	29 / h
Vehicles stopped on the crossing on red	6	20 / h
Red-light running	2	7 / h

Nobody was hurt in these 18 minutes, and none of it appears in any crash statistic.

User

The traffic engineer of a city district, who must decide where to change signals, markings or crossings.

Who pays

The city road-safety budget: a monthly subscription per existing camera. Second customer: insurers buying risk per intersection.

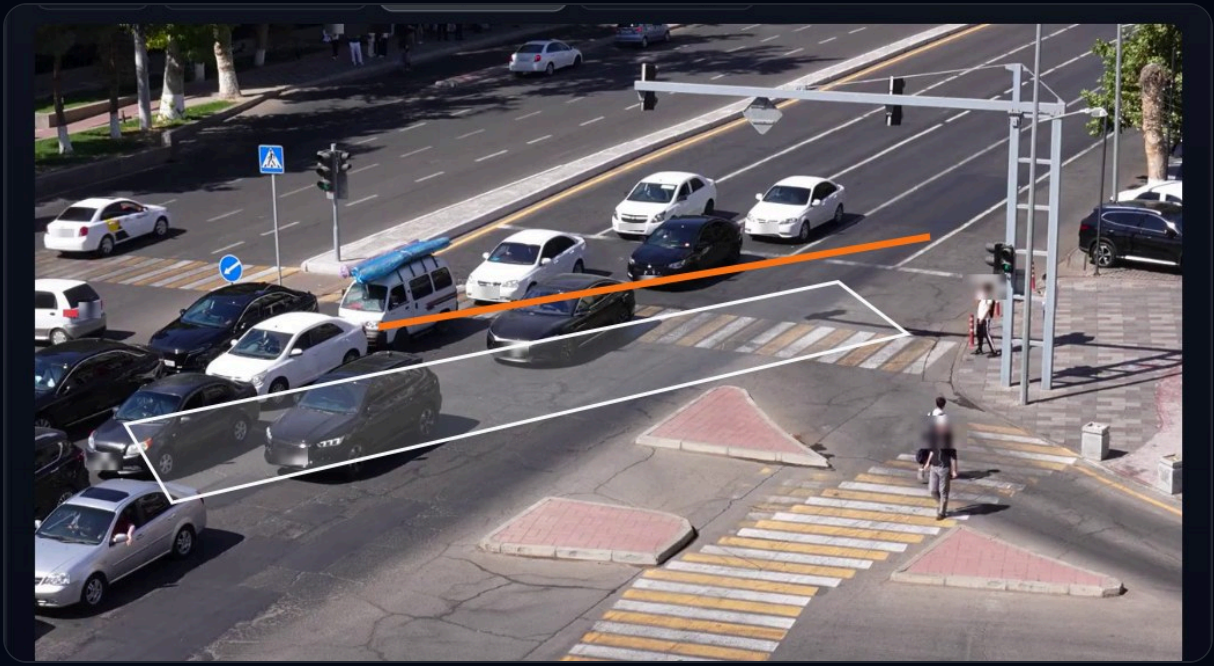
Today

Manual site surveys and crash reports. The cameras record, but nobody watches for near misses.

Counts: our own event labels of the organizer sample videos (elimination round)

PRODUCT

From one camera to a city-wide risk ranking



1 · Upload a video or photo. **2 · Optionally draw** road, crossing, stop line and light while it processes.



3 · Report + hotspot map + PDF for the engineer. **4 · City dashboard:** every camera ranked by risk, before/after comparison of a fix.

No zones drawn

Learns the road and lane directions from the traffic: wrong-way, near miss, stopped vehicles, slow traffic.

Zones drawn

Adds red light, stop line, pedestrians on the roadway and failure to yield.

Known camera

Recognised automatically and re-aligned, even if the camera shifted.

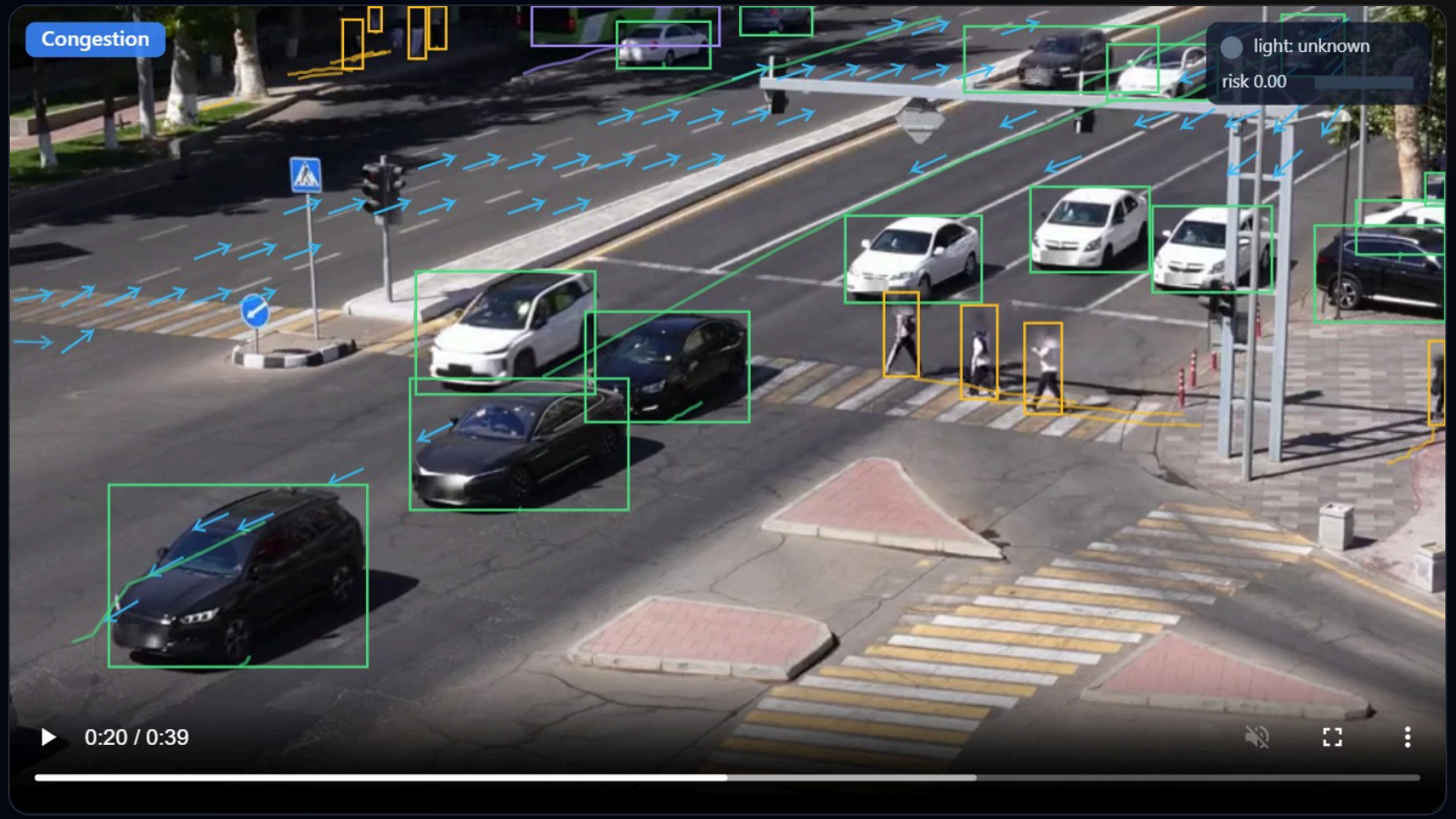
Screens from the live app

LIVE DEMO

Try it on your phone now

giyos1212-safejunction.static.hf.space

Frontend on Hugging Face; API on our RTX 3060 server; deployed by GitHub Actions



A camera it has never seen (zero-config): blue arrows are lane directions learned from the traffic in 40 s; boxes are tracked road users; heads and plates blurred.

Backup: recorded demo video in the repository

TECHNOLOGY

A system, not a call to someone else's model

1
Detect + track
Our YOLO26m fine-tune: 6 road-user classes + light state. ByteTrack. ~10 fps.

2
Scene
Drawn zones, or known camera re-aligned (ORB + RANSAC), or learned flow field: drivable area + lane directions.

3
Events + conflicts
10 interpretable rules (red light, yielding, wrong-way…) + **PET conflicts**, the measure city safety studies use.

4
Privacy
Plate detector run only inside each vehicle box + head blur, on the server.

5
Report + city
Hotspot map, PDF, recommendations; dashboard ranked per 1,000 pedestrians or by HSM Empirical Bayes.

Where it runs, how fast, what it costs

Inference	on-premise GPU server next to the city's video archive (ours: RTX 3060)
Speed, measured	**2× faster than real time at 10 fps · 1 photo in 0.5–0.7 s**
Capacity	5 fps loses no accuracy (ablation 0.828 → 0.827), so ≈ **4 cameras per GPU**
Night, rain, local vehicles	each camera auto-picks a detector: ours (Tashkent), our low-light fine-tune (synthetic night 0.39 → 0.92) or general COCO (night drone view: 5 → 43 confident detections in 6 frames); plates blurred, never read
Cost per camera	≈ **$12.7 / month infrastructure (server + power; assumptions on slide 8)**

Code: github.com/ubaydullayevgiyosiddin7-eng/doppix-safejunction

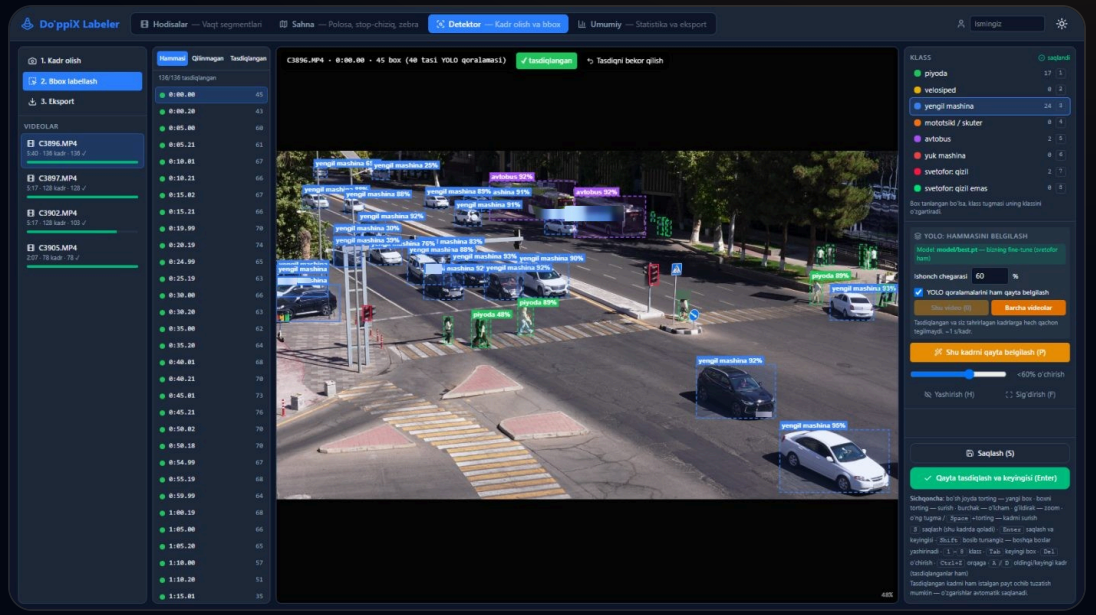
INNOVATION · WHAT IS HARD TO COPY

Our advantage is data, pipeline and local knowledge

Data: 12,990 boxes labeled by us
Tashkent traffic: Cobalts, Damas, articulated buses, our traffic lights. We built our own labeling tool (event timeline, scene editor, YOLO pre-labels) to make it.

Pipeline: zero-config scene learning
Lane directions learned per grid cell from traffic. A car is judged against the field with its own samples removed, so one wrong-way driver cannot teach the system its direction.

Industry methods, measured: PET conflicts, per-1,000 exposure, HSM Empirical Bayes
The metrics Miovision, Transoft/Brisk and Bellevue WA use, on any existing camera, checked against our own labels of 4 videos.



Our labeling tool: YOLO pre-labels, human review, export.

Replication test: the data and the measured rules take weeks, not a weekend

QUALITY ON TEST SETS WE BUILT

Honest numbers, including where it fails

What	Test set	Result
Detector	22 held-out frames, 1,386 boxes	**mAP50 0.978**
Event rules (zones)	our labels, 4 videos, 18 min	**Score A 0.83***
Wrong-way, zero-config	18 min real traffic	**0 false alarms / h**
Wrong-way, zero-config	60 real cars played backwards	**93 % caught**
Stopped / congestion, zero-config	our labels	**F1 0.0–0.67**
PET conflicts (new)	9 labeled failure-to-yield events	**7/9 have PET < 3 s**

* Optimistic: tuned on the same videos. Zero-config congestion fires on red-light queues; drawing a stop line fixes it.

Detector under synthetic weather and light (mAP50)

Condition	Daylight model	Low-light fine-tune (final)
Day	0.975 ✓	0.978
Fog	0.971 ✓	0.976
540p low bit-rate	0.881 ✓	0.857
Rain	0.793 ✓	0.967
Motion blur	0.712 ✓	0.959
Night	0.390	0.916 ✓

Night **0.39** → **0.92** after fine-tuning on synthetic night/rain/blur copies. It lowered the daylight event score (0.83 → 0.77), so **each camera auto-picks its detector** (ours, low-light or general COCO) from 6 sample frames (✓ = expected). Brightening frames was tried and rejected (0.27). Caveat: synthetic night; real night video is pilot month 1.

All methods and raw results: eval/ in the repository

BUDGET AND ROI

$12.4k pilot; profitable from 110 cameras

3-month pilot (8 cameras, 3 junctions)

Team: 3 people × 3 months × $900	**$8,100**
2 GPU servers (RTX 3060 class) × $1,200	**$2,400**
Labeling local footage (own tool)	**$1,500**
Power, hosting, network	**$400**
Licences: Ultralytics AGPL-3.0, open source	**$0**
Total	**$12,400**

Unit economics per camera per month

Price (assumption)	**$40**
Server $1,200 / 36 mo / 4 cameras	**$8.3**
Power 0.3 kW × 720 h × $0.08 / 4 cameras	**$4.3**
Gross margin	**$27.3 (68 %)**
Fixed costs: team $2,700 + other $300	**$3,000 / mo**

110
cameras to break even

300 → $5.2k
monthly profit

2.4 mo
pilot payback at 300 cameras

Every number except "4 cameras per GPU" (measured) is an assumption; each can be changed in the Business calculator on the live site. A closed-source commercial release needs an Ultralytics Enterprise licence or a permissively licensed detector.

Customer ROI is measured in the pilot: violations per hour before and after one change

PILOT IN THREE MONTHS · NEXT STEPS

Inside existing rules and processes

Month 1

Connect 8 cameras at 3 junctions

On-premise server at the city's video centre. Draw zones, start collecting, and record real night and rain footage.

Month 2

Label 2 weeks per site, retrain

Night and weather fine-tune; per-site precision checked by the engineer in the app.

Month 3

One change, measured

For example a leading pedestrian interval: violations per hour before and after, from the same camera, in the dashboard's before/after view.

Privacy and regulation

- Video never leaves the city server; faces and plates are blurred before any frame is shown.
- Only event metadata is stored; no identification of people or vehicles.
- Uploaded videos are deleted after 6 hours.

Next

- Night model from real footage
- Lane-level rules: illegal turns, solid lines
- Live RTSP streams instead of uploaded clips

TEAM DO’PPIX

Built during the final on our own elimination work



Eldor Musayev
Team captain · ML Engineer



Shohruh Gulmirodov
Web Developer



Ibodulla Obidjonov
Cybersecurity Engineer

Before the final (declared in the README): detector, tracker, zone rules for one intersection. During the final: any-camera mode, privacy blur, city report with hotspot map and PDF, city dashboard with before/after, low-light model with day/night routing, robustness tests, CI/CD. Git history shows both. Built with an AI coding assistant; open source: Ultralytics, PyTorch, OpenCV, FastAPI, React.

Try it: giyos1212-safejunction.static.hf.space